PRODUCT DATASHEET · SKU PIA24028P

Brain Rhythms In. Real-World Actions Out.

A hands-on learning kit that links the Octopus 16 wireless EEG board to an Arduino starter board. Students read their own brain signals, stack simple block programs and make lights, motors, sounds and a screen respond — all in one browser tab at octobridge.pieeg.com, with no install and no compiling.



16	15	26	5	0
EEG channels	Sensors & modules	Lessons in the manual	Block levels	Installs needed

01 WHAT IS THE OCTOBIO KIT?

	Brain side	Room side
Device	Octopus 16 EEG board (PiEEG)	All-in-One Starter Kit for Arduino (Elecrow SEA0004T)
Link to the browser	Web Bluetooth	Web Serial over USB, 115200 baud
Role	Reads EEG; the browser turns it into band power	Senses the room and acts on it
Firmware	OctoBio / Octopus-16 firmware, flashed from the browser	OctoBio Kit firmware, flashed from the browser

The All-in-One Starter Kit for Arduino (SEA0004T) is designed and manufactured by Elecrow. The Octopus 16, the OctoBio firmware for both boards and the octoBridge web lab are by PiEEG.

02 SYSTEM DIAGRAM

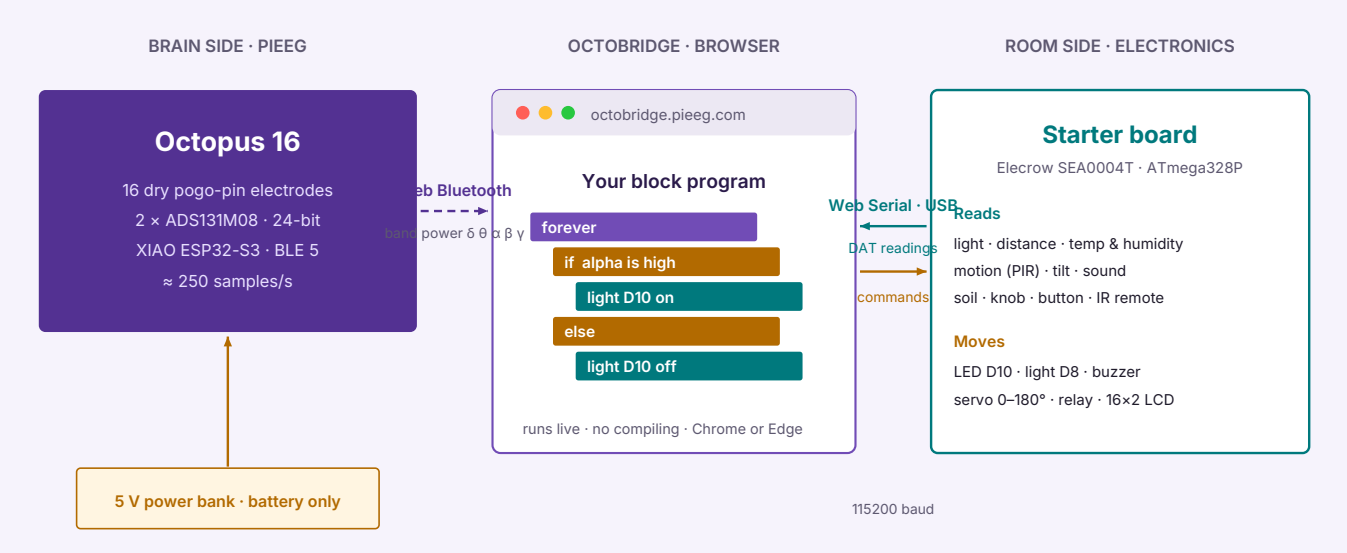


Figure 1 — Both devices stay live at once. The block program runs in the browser and sends commands to the starter board.

03 TECHNICAL SPECIFICATIONS

OCTOPUS 16 EEG BOARD · PIEEG	
Electrodes	16 dry spring-loaded pogo pins + reference + ground in one Ø 26 mm board
Converters	2 × Texas Instruments ADS131M08, 24-bit, gain 4
Controller / radio	Seeed Studio XIAO ESP32-S3 · Bluetooth LE · name Octopus-16-XXXX
Sample rate	≈ 250 SPS (≈ 244 SPS on boards with an 8.000 MHz ADC clock)
Power	Small 5 V power bank · battery only while worn
STARTER BOARD · ELECROW SEA0004T	
Controller	ATmega328P, Arduino Uno compatible · 32 KB flash, 2 KB RAM, 1 KB EEPROM
USB	USB-C with CH340C USB-to-UART · 115200 baud to octoBridge
Sensors (10)	Light BH1750 · ultrasonic distance · DHT20 temperature & humidity · PIR · 6-axis IMU · sound · soil moisture · slide potentiometer · button · IR receiver
Outputs (6)	Red LED D10 · light D8 · buzzer · relay · servo 0–180° · 16×2 blue-backlit LCD
Expansion	6 Crowtail ports: 3 × I/O (A3, A6, D11), 2 × I²C, 1 × UART
Power	5 V over USB-C · 42 mA idle, 138 mA with everything on (0.21–0.69 W)
Mechanical	195 × 170 × 46 mm in its PP carrying case · 380 g
OCTOBRIDGE WEB LAB · PIEEG	
Platform	Desktop Chrome or Edge · octobridge.pieeg.com · no account, no install
Programming	Blocks in 5 levels (Story → Raw), run live without compiling; Write page shows the same program as text
EEG values	Relative band power δ θ α β γ · focus β/(β+θ) · relax α/(α+β) · RMS µV · "EEG ok" 2–200 µV

04 WHAT STUDENTS LEARN

Subject	Topics
Biology	Brain rhythms (δ θ α β γ), why alpha rises when the eyes close, the artifacts that blinks and jaw clenches leave in the signal
Electronics	Sensors and actuators, analog vs digital, I²C, serial communication, units and ranges
Programming	if / else, loops, thresholds, mapping one range onto another, combining two signals into one decision

Part A · starter board only	Lessons
LED, button, screen, moisture, street light	1–6
Distance, obstacle alarm, plant watering, slider	7–10
Weather, servo, follow the knob, IR remote	11–15
Automatic door, night light, sound, tilt, corridor	16–20

Part B · with the Octopus 16	Lesson
Eyes closed, light on	21
Focus meter	22
Quiet room (alpha + no motion)	23
Blink is not a thought	24
Brain dashboard · mind door	25–26

05 ORDERING INFORMATION

Product	OctoBio Kit — brain-to-hardware learning kit
SKU	PIA24028P
Price	USD 370
Purchase	pieeg.com/store/octobio-kit · Elecrow (official manufacturing partner)
Repository	github.com/pieeg-club/OctoBio-kit — user manual, Octo Studio, images

Included	
1 × Octopus 16	With XIAO ESP32-S3 and BLE antenna
1 × Starter Kit for Arduino	Elecrow SEA0004T, in its case
Software	octoBridge web lab and firmware for both boards

Required, not included	
Small 5 V power bank	For the Octopus 16
Computer with Bluetooth	Desktop Chrome or Edge
USB data cables	USB-C; charge-only cables do not work

WARNING Battery only on the head. Never mains on the relay.

The Octopus 16 has no isolation from its power supply. While it is on a head it runs from a small 5 V power bank only — never plugged into a computer or a wall charger. USB is for flashing only, with the board off the head. Never connect the relay to mains voltage. The OctoBio Kit is not a medical device.

06 RELATED DOCUMENTS

OB-MD-002 Mechanical & Layout Drawing	**OB-PCB-003** Hardware & Manufacturing Specification	**OB-PIN-004** Pinout & Connection Guide
OB-QSG-005 Quick Start & Safety Guide	**OB-SW-006** Software & Protocol Guide	